Online Age Verification: AI as a Solution?

By Jessica Eynard, Associate Professor in Law, University of Toulouse Capitole

In this lecture, Eynard unpacks the challenge of online age verification. She looks at the legal and technical frameworks which currently exist to protect minors, exploring how regulations by the European Union manifest in the national contexts of France and Germany. Through this exploration, she highlights the benefits and limitations of the current AI solutions. Eynard concludes her mondule with an outline of future considerations to govern online AI-powered age verification systems.

Lecture Transcript

0:02 Good afternoon, everybody. My name is Jessica Eynard. And it is my honor to be here with you to propose this module in the context of the Global AI Ethics Consortium, and the Institute for Ethics in Artificial Intelligence, and I would like to thank the organizer of this Consortium Institute for having invited me to make this module.

0:38 So maybe before beginning, I can just present myself in a few words. So, my name is Jessica Eynard, as I said, I'm an Associate Professor in Law in Toulouse at the University of Toulouse in France. I'm specialized in data protection law, in digital identity law, and I work on artificial intelligence regulation as well. And I need these three specializations to present the topic that I'm going to present to you now, which deals with online age verification. And the question will be to determine if artificial intelligence can be a good solution to verify the age online. Just for you to know that this contribution comes from another contribution I made in a book I co-edited with the Professor Céline Castets-Renard, who is a professor at the University of Ottawa in Canada. And so if you want to know more about this, and about artificial intelligence, you can refer to this book.

2:11 So if I go back now to my topic. Why this topic? This topic has an origin. It comes from the problems that arise from the way Internet is built. Because when Internet has been built, the purpose was to identify or the need was to identify the machine, the computer, and not to identify the user of the machine. And that's the reason why on the internet, nobody knows that you are a dog. But sometimes, for legal reasons, but not just for legal reasons, you need to know who is behind a machine, who is the user. You need to identify him or her. Or at least you need to know some attributes about these users. Because some websites can be accessed just by some groups, and in particular, minors are prohibited to access to some websites.

3:28 So is there a regulation on that topic? I think I can definitely say yes, at the European level, but also at the national level. Concerning the European level, I made a focus just on three texts, that seems for me to be more important on that topic. You have the famous GDPR, General Data Protection Regulation, dated 2016, which creates, which provides some rules concerning children, minors, to protect them. You also have a second text is the Audio Media Services Directive, dated 2018, which (sorry) which creates some obligations for a group of providers in order to protect minors. And more recently, you get the Digital Services Act, dated 2022, which still provides obligations for some, some platforms, and in particular, for very large online platforms and very large online search engines. You can find some disposition concerning the systemic risks and the obligation for those actors to take measures to mitigate this risk. And a kind of measure will be to take measures to verify the age of the user, and to put in place some tools for parental control.

5:42 At the European level, national level, you find as well, many, many laws. As far as France is concerned, the main text is the law of July 30, 2020, from which you can deduce that pornographic content must be inaccessible to minors under 18. And that just a disclaimer is, it's not enough. So you have to take more measures. And the organization is fined and the website can be blocked or delisted from search engine results. So we have many, many regulations. And they all provide the same two things that are measures to be taken to verify age and tools for border control. You always find these two.

6:45 Okay, we get the legal tool, but what can we observe? We can observe that that doesn't work. That doesn't work because it's sufficient to say, to click on "I want to view the content", or "I am above 18", to access to content. And inquiries proved that it's easy for minors to access illicit content for them. And in particular, pornography content. An inquiry in France, showed that by age 12, nearly one child in three has already been exposed to pornographic content. And over 82% minors have been exposed to pornographic content. So we got the legal tools, but clearly they don't work. Why so? Why so because we're trying to conciliate different interests, and at least three interests. The first one is the minor protection. So the solution adopted should be efficient, we should be able to rely on these tools. Second, interest. Privacy and personal data should be protected. So it's not easy to protect personal data and privacy and to be efficient in the same time. Because most of the time you need data to be efficient, reliable. And the third interest is that if it's possible, it would be very, very great if the solution adopted could be user friendly. That is to say that if someone must click 40 times before accessing the website, this is not so user friendly, and maybe a person will stop clicking and so won't access to the website before it.

9:00 So not so easy to conciliate these interests. We have solutions under consideration. You can think about the ID card, the electronic ID card. You can think about record or facial recognition. You can think about using credit cards, because we can imagine that only people who are not minors have credit cards allowing to access illicit content. We can think about the blockchain. We can think about using trusted third party or a parental control. We have solutions. But what I propose here is too focused on solutions that are using AI to verify the age online. Two main techniques can be used to estimate the age online thanks to artificial intelligence. The first one consists in estimating an approximate age, based on an analysis of the user's online behavior. That means that we'll have a look on what type of content is consulted and published by the user, what kind of passwords are used, what is the environment, the global environment of the users. And thanks to that different information, it will be possible to deduce an approximate age. And the second technique consists in estimating an approximate age from a photo or a short video of the user.

11:04 So here are the techniques. And we all have the same regulation, at least at the European level. But what we can observe is that these regulations are not interpreted the same way for these solutions by the different national authority. And to prove that we can take the example of two national authorities. The first one is the French National Authority, which name is the CNIL for Commission nationale de l'informatique et des libertés. And the second one is the German authority, which name is KJM for Die Kommission für Jugendmedienschutz. And we will see that they don't retain, they don't have the same solution at the end. If we consider first German, the German authority, we can see that this authority concludes positive assessment of over 100 technical solutions for age verification. The main part of those solutions remain on an electronic ID card, plus a photo or a selfie. But recently, the German authority validates solutions based on facial estimates. So we don't need the identity card anymore, or the passport anymore. We just try to estimate your age, thanks to your face. And so thanks to biometric characteristics.

13:03 And as it's not so easy to determine whether a person is minor or not, and especially when the person is 17, or 18, or 19. These solutions used results at 23 years old. I mean that, for a user to access the website, the solution should conclude that this user is at least 23 years. So that creates something very noticeable for the users between 18 and 23, because these users can be prohibited to access a website even if they are the age required to access those websites. So a special procedure or procedure should be put in place for the person and it's for them, it's harder to access the website. And I'm wondering if in a way, we are not living kind of a shift of minority from 18 to 23. So the future will say.

14:34 For French authority, the reality is not exactly the same, because we don't have as many solutions validated as in Germany. In a first decision, the French authorities decided that no technical solutions were ideal. And they all have their default, their drawbacks. And for example, the CNIL analyzed the browsing history. And for this authority using this to estimate age was disproportionate regarding the risk for rights and freedoms, and it was not really reliable. Concerning the facial recognition or the identification thanks to a photo of biometric, the French authority decided not to validate it. As for CNIL, it was disproportionate regarding the rights and liberty and freedom and concerning the risk for these rights and freedom. So that was our first position, but then this position evolved because it was necessary to find solutions for the websites, which needed to verify the age online. And so in a position of 2022, CNIL validated some solutions. And in particular, they validated the average age estimation based on facial analysis. So without identification, we just estimate your age without knowing who you are, and without knowing your name or your first name. And what the CNIL decided at that moment is that the solutions adopted should be certified and deployed by a trusted third party complying with precise specifications.

17:04 And I think that we are here touching at something very important, because we are trying to have a high level of guarantee that means for the solution to be reliable. So that's what is required by a law, by lawmakers. But at the same time, we never say how to do so. How to reach this high level of guarantee of security. And that's the problem. Because no solution seems to be ideal. If we have a look on some texts, we can find some beginnings of answer. If you look at the DSA say for example, the Digital Services Act, it's written in it that the European Commission should publish guidelines to help providers to protect the users' rights, in particular the minors' rights. In the French law, it's written that special authority which is ARCOM should adapt guidelines with the enlightenment of the of the CNIL French authority concerning data protection, personal data protection.

18:46 So we have some little things in the law but clearly that it's not enough because we need technical specification and that's exactly what was proposed by the European Commission in communication dated 2022 Because in this communication, the European Commission provided to launch a standardization request for a European standard online age checking verification. And the European Commission would like an EU wide digital proof for age that is based on date of birth, so that must be reliable at the end. So we need the standardization, this specification to be adapted to go on with this difficulty of online age verification.

20:03 At the French level as AI and as other solutions, were not either reliable or either protective enough of rights and freedom. The French legislators decided to focus on another solution, which is based on a process with double anonymity. So, when someone, when a user wants to access a website, this website should ask the users to give the proof of him not to be minor, for example. And to do so the user can rely on a trusted third party who will have reliable information coming from the administration, from a bank, or from another another way. And so this trusted third party will just send an assent to the user saying “this user is minor or is not minor”. And this assent will be sent to the website for the website to know if the user can access another website. But at the end, this scheme allows a double anonymity because the website will never know the identity of the user. And the trusted third party will never know which kind of website the user wants access to. So this is what is put in place in France now. We've got other solutions based on ID card as well. But this is what is put at least. This has drawbacks, this doesn't work. Or as in maybe we can say that it's even easy not to use this process that means to bypass this process to access a website, even if you are a minor.

22:23 A solution that is to come at the European level and that I want to mention at the end of this module is the wallets, the European Digital Identity wallet, which is in the proposal for amending the eIDAS Regulation. And this wallet could be a very interesting tool to verify the age online, because it's very secure. There's a triple certification, because it's based on either official identity documents. So it's reliable. The guarantee is at a high level, and it respects the GDPR. And in particular the principle of minimization. So we have to keep an eye on this wallet to see how it will be implemented in the future to be sure that rights, freedoms are still protected. But this could be a very good tool for the future.

23:39 To conclude with this module, maybe what we can say now is that the online age verification framework is under construction. We have to think this framework. And I think that we have to be vigilant, because behind this question, we have the question of being able to be on the internet when being anonymous, when staying anonymous. What do we want? Maybe the boundary should be drawn to define what we want. More identification or more anonymity? That's the question I think, that is important behind all of this. I really thank you to have looked to this module, to have listened to me. I hope that it has been interesting for you to learn about online verification of age, and I hope that you learned more, you learnt a lot (sorry) when being with me for this few minutes. Thank you.